Task 1A: Initial Data Analysis

Sofia French

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Overview

This document summarises the steps completed for Task 1A in the data science project. The focus has been on understanding how perceived wine quality varies between red and white varieties, through data analysis: specifically, distribution plots.

Quality Distribution – Red Wine

The first analysis targeted the distribution of the `quality` variable in the red wine dataset. The quality values range from 3 to 8, with most samples concentrated around 5 and 6.

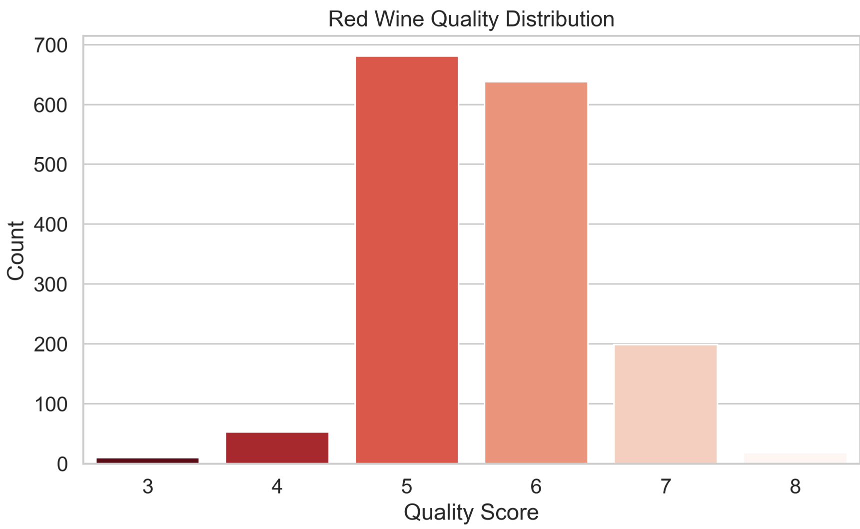

Figure 1: Distribution of Quality Scores in Red Wine Dataset

Observation: The majority of red wine samples are rated as quality 5 and 6, which together make up the bulk of the dataset. There is a sharp drop for higher quality scores (7 and 8), and very few samples with scores below 4.

Quality Distribution – White Wine

The same analysis was performed on the white wine dataset.

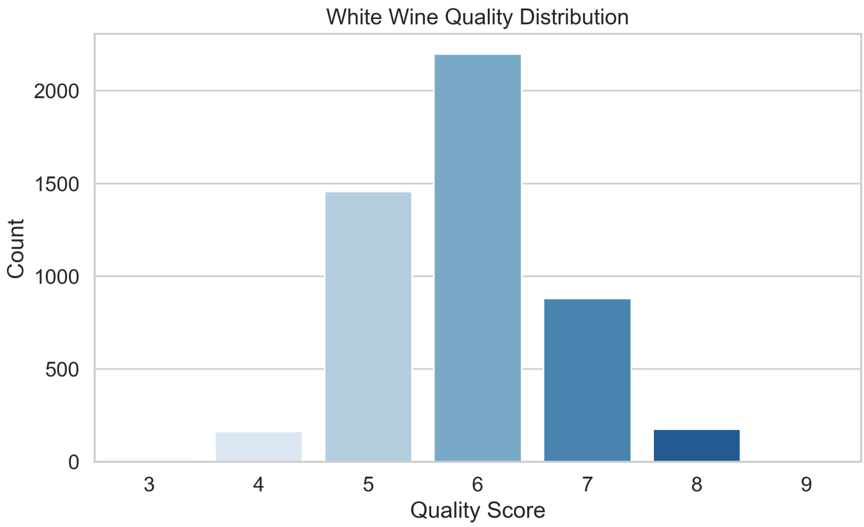

Figure 2: Distribution of Quality Scores in White Wine Dataset

Observation: Like red wines, most white wines fall into the 5–6 quality range. However, white wines show a broader spread — with more wines rated as 7 and 8 compared to red. There's also a noticeable number of white wines rated as 4, which wasn't as prominent in the red wine data.

Red vs. White Comparison

To better understand how red and white wines compare, both quality distributions were plotted together (see Figure 3). It is important to note that the white wine dataset contains over three times as many samples as the red wine dataset (4,898 vs. 1,599), which impacts the visual scale and interpretation.

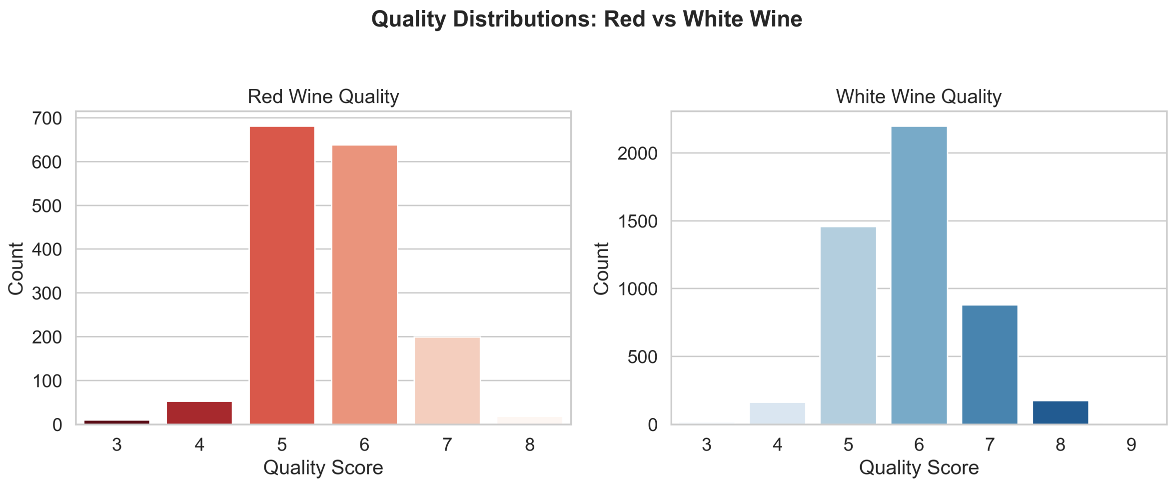

Figure 3: Side-by-Side Distribution of Red and White Wine Quality

Interpretation:

- **Red Wines:** Quality scores are tightly clustered around 5 and 6, with relatively few samples scoring outside that range. This suggests consistency, but also limited variation.
- **White Wines:** While also peaking at 5 and 6, the distribution is notably wider. A significant number of samples score 7, 8, or even 9 — which are absent from the red wine data. There are also more wines scoring 4.
- **Sample Size Consideration:** The broader spread in the white wine distribution is partly due to the **larger sample size**, which introduces more variability and makes higher and lower quality scores more visible.
- **Overall Implication:** While both types of wine tend to cluster around mid-level quality, white wine exhibits a **greater diversity** in perceived quality, which may point to differences in grape variety, fermentation control, or batch variation.

These early observations will guide further investigation into physicochemical factors that may explain why some wines — especially white — achieve higher sensory scores.